Homework 5

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Disclaimer: Generative AI was used on problem 3 for understanding the core concepts involved and on problem 6 for help computing the conjugacy classes of D_8.

1. Let Q_8 be the Quaternion group. Recall that $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, $ji = -k$, $kj = -i$, and $ik = -j$.

(a) Find all composition series of Q_8 and list the composition factors.

(b) Illustrate Cayley's Theorem applied to Q_8 by identifying an isomorphic copy of Q_8 as a subgroup of S_8.

(c) Prove[1] that Q_8 is not isomorphic to a subgroup of S_n for any $n < 8$.

2. Compute the orders of the conjugacy classes of S_6, and use this to compute the order of the centralizer of each type of permutation in S_6. Also, use Problem 8 on the previous homework to determine the size of the conjugacy classes of A_6. Finally, determine the class equation for both S_6 and A_6.

3. Let G be a group of order $3825 = 3^2 \cdot 5^2 \cdot 17$. Prove that if H is a normal subgroup of order 17 in G, then H is contained in the center of G.

Proof. We let G act on H by conjugation, which induces a homomorphism $\varphi : G \to \mathrm{Aut}(H)$. Note that $\mathrm{Aut}(H) = 16$ since H must be cyclic (because 17 is prime). The image of this homomorphism must divide both $|G|$ and $|\mathrm{Aut}_H|$, but 3825 and 16 are relatively prime, so $\mathrm{im}\,\varphi$ must be trivial. Thus $H \subset Z(G)$. $\square$

4. Prove that if K is a characteristic subgroup of H and H is a characteristic subgroup of G, then K is a characteristic subgroup of G. Also show that if K is a characteristic subgroup of H and H is normal in G, then K is normal in G.

Proof. Given an automorphism φ of G, $\varphi|H$ is also an automorphism, and so $(\varphi|H)|K = \varphi|K$ is an automorphism. Thus K is characteristic in G.

Now consider conjugation by some element $g \in G$. This induces an automorphism on G, which restricts to an automorphism on H because H is normal. Since K is characteristic in H, the restriction to K is again an automorphism, and thus K is normal in G. $\square$

[1] Hint: Prove that if Q_8 acts on a set X with $|X| \leq 7$ and $x \in X$, then $\mathrm{Stab}_G(x)$ must contain $\langle -1 \rangle$.

5. Let G be a group, and H a proper subgroup of G such that $|G| > |G : H|!$. Prove that there must exist a nontrivial normal subgroup of G contained in H, so that G is not simple.

6. Determine whether or not each of the following class equations are possible for a group of order 10. If it is possible, provide a group that realizes it. (Here, I am writing the elements of the center as singleton conjugacy classes).

(a) $1 + 1 + 1 + 1 + 1 + 5$

(b) $1 + 1 + 1 + 2 + 5$

(c) $1 + 2 + 2 + 5$

(d) $1 + 2 + 3 + 4$

(e) $1 + 1 + 2 + 2 + 2 + 2$

Proof. Let G be the general notation for a candidate group of order 10.

(a) We can see that $|Z(G)| = 5$, and since the only other term is 5, there is one conjugacy class with size 5. Then $|C_G(x)| = 10/5 = 2$, but this is impossible since $Z(G)$ is a subgroup of $C_G(x)$. Thus, this is not the class equation of any group. This reasoning will be abbreviated for future items.

(b) Since $|Z(G)| = 3$ and $|C_G(x)| = 5$ for some x, this is not the class equation of a group.

(c) D_{10} satisfies this class equation since the conjugacy classes are $\{1\}$, $\{r, r^4\}$, $\{r^2, r^3\}$, $\{s, rs, r^2s, r^3s, r^4s\}$.

(d) Since 3 does not divide 10, this is not the class equation of a group.

(e) Since $|Z(G)| = 2$ and $|C_G(x)| = 5$ for some x, this is not the class equation of a group.

□

7. Suppose that G is a group with class equation $1 + 4 + 5 + 5 + 5$.

(a) Does G have a subgroup of order 5? If so, is it a normal subgroup?

(b) Does G have a subgroup of order 4? If so, is it a normal subgroup?

Proof. Let x be an element of the conjugacy class of order 4, and let y be an element of one of the conjugacy classes of order 5.

(a) Yes, $C_G(x)$ is such a subgroup, since $|C_G(x)| = |G|/4 = 20/4 = 5$. The union of conjugacy classes is normal if it is a subgroup, and indeed we have a subgroup in the union of the trivial conjugacy class and the order-4 conjugacy class, so we have a normal subgroup of order 5.

(b) Yes, $C_G(y)$ is such a subgroup, since $|C_G(y)| = |G|/5 = 20/5 = 4$. Since 4 cannot be written as a combination of 1 with any other of the orders of conjugacy classes, it cannot be a normal subgroup.

$\square$

8. Let G be a nonabelian group of order p^3 with p a prime integer. Determine $|Z(G)|$. Next, determine $|C_G(x)|$ for $x \notin Z(G)$. Finally, determine the class equation of G using this information.